

NineRec: A Benchmark Dataset Suite for Evaluating Transferable Recommendation

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Abstract—Large foundational models, through upstream pre-training and downstream fine-tuning, have achieved immense success in the broad AI community due to improved model performance and significant reductions in repetitive engineering. By contrast, the transferable one-for-all models in the recommender system field, referred to as TransRec, have made limited progress. The development of TransRec has encountered multiple challenges, among which the lack of large-scale, high-quality transfer learning recommendation dataset and benchmark suites is one of the biggest obstacles. To this end, we introduce NineRec, a TransRec dataset suite that comprises a large-scale source domain recommendation dataset and nine diverse target domain recommendation datasets. Each item in NineRec is accompanied by a descriptive text and a high-resolution cover image. Leveraging NineRec, we enable the implementation of TransRec models by learning from raw multimodal features instead of relying solely on pre-extracted off-the-shelf features. Finally, we present robust TransRec benchmark results with several classical network architectures, providing valuable insights into the field.

Index Terms—Benchmark, dataset, fine-tuning, modality-based recommendation, pre-training, transferable recommendation.

I. INTRODUCTION

RECOMMENDER system (RS) models play a crucial role in predicting user preferences for unseen items based on their previous interactions. These highly successful models have found wide-ranging applications, such as in advertising systems, e-commerce websites, search engines, and streaming services. In the past few decades, extensive research has been conducted on both content-based [1] and collaborative filtering [2], [3], [4] recommendation models. Among these approaches, ID-based collaborative filtering models (known as IDRec), which leverage

unique IDs to represent users and items have dominated the RS field for over 10 years.

Meanwhile, the IDRec paradigm encounters several key bottlenecks due to its inherent characteristics. First, IDRec struggles to handle cold-start scenarios because new userIDs and itemIDs cannot be effectively trained before being deployed in live environments. Second, the design philosophy of IDRec diverges from the fundamental principle of modern “foundation” models [5] in the deep learning community, which emphasizes the adaptability of pre-trained parameters to multiple downstream tasks. This is because IDRec usually requires either shared data or overlapped IDs (i.e., userIDs and itemIDs) to realize cross-domain recommendation [6], [7], [8], [9], [10], [11]. However, achieving such cross-domain recommendation often proves impractical due to concerns related to data privacy and overlap rates between different systems. For instance, platforms like TikTok may not share their userIDs or videoIDs with platforms like YouTube.

To overcome these limitations, an intuitive approach is to abandon the use of userID and itemID features, particularly itemID.¹ Instead, we can leverage the multimodal content of items to represent them [10], [12], [13], [14], [15], [16]. We refer to this approach as MoRec [12]. For example, if the item is a news article or text, we can utilize BERT [17] or RoBERTa [18] to represent it. If the item is an image, we can employ ResNet [19] or Vision Transformer (ViT) [20] to represent it. By representing items with modality features, recommendation models can naturally possess transfer learning capabilities across domains and systems. This paradigm, called TransRec, shares similarities with universal models in natural language processing (NLP) & computer vision (CV).

However, TransRec models have received less attention and success than NLP and CV. So far, the RS community does not have a downloadable TransRec model, such as on platforms like HuggingFace, whose pre-trained parameters can be directly applied to other recommendation datasets, akin to the usage of BERT in NLP. There are several challenges for the successful deployment of TransRec models in practical applications. One major challenge is the strong establishment and dominance of the IDRec paradigm, which has represented state-of-the-art baselines for over a decade, especially in non-cold start scenarios. TransRec or MoRec that rely solely on multimodal features

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To facilitate further research, we will release our code, datasets, benchmarks, and leaderboards at <https://github.com/westlake-repl/NineRec>.

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¹This is because the userID can be represented by the itemIDs that the user has interacted with, as seen in most sequential recommendation models.

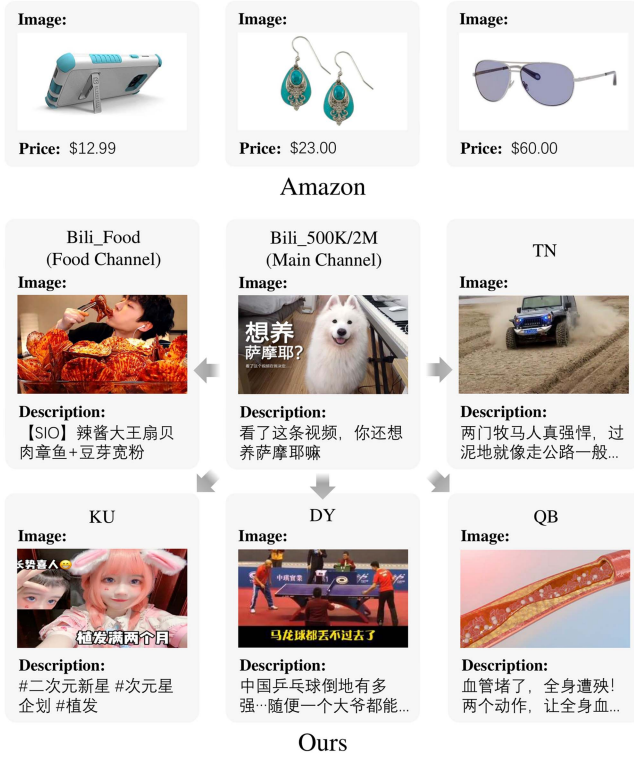


Fig. 1. *Image cases of NineRec versus Amazon.* (a) Compared to Amazon, the images in NineRec are more abstract and semantically enriched; (b) NineRec supports cross-platform recommendation; Bili, TN, KU, DY, QB are different recommender systems; and (c) User intent in Amazon is largely influenced by item price, which cannot be achieved by learning only appearance or visual features.

often struggle to outperform these IDRec models,² particularly in past years when highly expressive modality encoders, such as large BERT or top-performing ViT, were not yet available. This situation has seen some progress in recent months, as evidenced by recent literature [12], [16], which confirms that even in non-cold start and warm item scenarios, itemID features can be replaced with an advanced multimodal encoder.

Another challenge for the TransRec paradigm is the scarcity of large-scale multimodal pre-trained recommendation datasets and diverse downstream datasets. While Microsoft provides MIND [22], a high-quality news recommendation dataset, it lacks diverse downstream datasets for evaluation and does not include raw image features. Several e-commerce datasets, such as Amazon,³ Yelp,⁴ and GEST [23], can provide raw image features, but the items in these datasets often revolve around simple objects (as depicted in Fig. 1) or have limited visual diversity,⁵ making them less suitable as pre-training datasets for general or semantically richer images. More importantly, these datasets are intuitively less optimal for studying pure modality (visual or textual) recommendations as user intent in

e-commerce datasets is heavily influenced by other factors, such as price, sales, brand, location, and most importantly, the user's actual purchase needs.

In this paper, our primary goal is to solve the dataset challenges for the community, and subsequently provide reliable benchmarks. Specifically, we introduce NineRec, a TransRec dataset collection consisting of a very large source domain dataset (with 2 million users, 144 thousand items, and 24 million user-item interactions) and nine diverse target domain datasets (including five from the same platform with different scenarios and four from different platforms). Each item is represented by an original descriptive text and a high-resolution cover image. To the best of our knowledge, NineRec is the first large-scale and highly diverse datasets for streaming content recommendation, encompassing various types of raw content, including short videos, news, and images. One distinctive characteristic of our NineRec datasets is that user watching intent in streaming media can be primarily inferred from the visual appearance of items, with minimal influence from non-visual factors like price in e-commerce recommendation datasets or distance in location recommendation datasets. From this perspective, NineRec is a more ideal dataset for studying multimodal content-focused recommendation. We then report several representative TransRec baselines for visual and text recommendation tasks on the source dataset and nine target datasets by replacing ID embeddings with advanced modality encoders. Our rigorous empirical studies on NineRec have uncovered several interesting findings. To facilitate future research, we release our code, datasets, benchmarks, and leaderboard. Beyond this, we envision NineRec as a useful dataset for the NLP & CV researchers, who can use recommendation as a downstream task to evaluate the generality of new image/text encoders. Given this, NineRec helps unify the fields of RS, NLP & CV.

II. NINEREC DATASET SUITE

A. Dataset Summary

To facilitate the TransRec research, we curate a suite of benchmark datasets which comprise a large-scale source domain dataset from Bili and nine different downstream target domain datasets, namely, Bili_Food, Bili_Dance, Bili_Movie, Bili_Cartoon, Bili_Music, QB, TN, KU and DY.⁶ Bili, KU and DY are three most famous short-video RS platforms in China, where each item is a short-video,⁷ while TN and QB are two large streaming recommendation platforms where an item can be either a news article, short-video or an advertisement. Each item in all of the above datasets contains a textual description and an image cover. Each positive user-item interaction is either a thumb-up or a comment, which is a strong signal for user preference. Note that we do not retain the contents of comments as we consider the textual description (i.e. title) of an item to be more representative than comments or reviews.

We provide two source datasets: Bili_500K and Bili_2M, where 500K and 2M stand for 500 thousand and 2 million users,

²A typical example is that to date, except for the recent M6-Rec [21], almost no real-world online recommender systems have explicitly claimed that they have completely abandoned the itemID feature.

³[Online]. Available: <http://snap.stanford.edu/data/web-Amazon.html>

⁴[Online]. Available: <https://www.yelp.com/dataset/download>

⁵For example, item images in Yelp and Gest are mostly about food and restaurants.

⁶Bili: <https://www.bilibili.com/>; QB: <https://browser.qq.com/>; TN: <https://news.qq.com/>; KU: <https://www.kuaishou.com/new-reco>; DY: <https://www.douyin.com/>.

⁷Videos with play time longer than 10 minutes are not collected.

TABLE I
DATASETS

Dataset	#Users	#Items	#Actions.	Sparsity	Image Resolution	#Description (Chinese)	#Description (English)
Bili_500K	500,000	138,033	7,845,805	99.99%	1920x1080	21.91	19.28
Bili_2M	2,000,000	144,146	24,497,157	99.99%	1920x1080	21.90	19.27
Bili_Food	6,549	1,579	39,740	99.62%	1920x1080	24.91	22.15
Bili_Dance	10,715	2,307	83,392	99.66%	1920x1080	19.62	18.00
Bili_Movie	16,525	3,509	115,576	99.80%	1920x1080	24.93	21.99
Bili_Cartoon	30,300	4,724	215,443	99.85%	1920x1080	19.26	16.20
Bili_Music	50,664	6,038	360,177	99.88%	1920x1080	21.60	19.70
KU	2,034	5,370	18,519	99.83%	360x640	25.00	19.13
QB	17,722	6,121	133,664	99.88%	496x280	25.26	20.74
TN	20,211	3,334	122,576	99.82%	496x280	26.76	22.12
DY	20,398	8,299	139,834	99.92%	300x400	34.46	26.92

The sequential order of actions are retained. #Description (Chinese) and #Description (English) is the average word length of text description in Chinese and English, respectively.

respectively. Bili_500K is a subset of Bili_2M. Their collection strategies are similar and will be given in the following subsection. The user-item interactions for the source datasets were collected from both the main channel and 20 vertical channels, resulting in a highly diverse range of item categories. In contrast, the Bili_* datasets in the target domain were collected from five vertical channels (excluding those from the source datasets) of the Bili website, where the items on each channel page are mostly from the same category. For example, items in Bili_Food are mainly about food and cooking, while items in Bili_Music are music videos. Bili_2M and Bili_* have no overlapped items or user-item interactions. There might be a small number of users who visited both the main and these vertical channels. But we do not consider the overlapping users as it is not our focus.

B. Dataset Construction & Analysis

The data collection process lasted approximately 10 months from September 2021 to July 2022. Taking Bili source as an example, we collected short videos from more than 20 channels (including main channels with various categories). By frequently requesting the webpage, we could collect about 1000–2000 videos per channel. We then went to the pages of all these videos, which often contained many links to other videos. In each page, we randomly selected 3–5 videos. We did this many times. Then, we merged all videos and removed duplicates. As for user feedback, we went through all pages of collected videos and collected the public comments (including the bullet-screen comment) for each video, and ensured that each video has at most 3,500 user comments. We crawled the comment data page by page, and the more comments we collect, the longer it will take. We only recorded the latest pair of user-video interactions even though the user might commented on a video multiple times.

For the source dataset, we first got Bili_500K after a few months of data collection, then aggregated all the data and removed users with less than 10 behaviors. We then proceeded to crawl more data over several months and aggregated all existing data, but only removed users with less than 5 behaviors, resulting in Bili_2M. Due to the slow collection time of Bili_2M, we conducted the major experiments on Bili_500K. Similarly, we

collected data from 5 vertical channels of Bili and four other platforms, i.e. QB, TN, KU and DY. For these downstream datasets, we maintain the same data collection procedure. In this paper, we only keep the cover image and title description to represent an article or a short-video instead of the news contents or original videos. After basic processing, we recruited five students who manually checked the quality of images and text and removed about 1% of items that were poor quality (e.g. images with just black background, image-text mismatches, overly sensational text descriptions, etc.). We preserve main properties of these datasets without more pre-processing since they might be important for other research. The statistics of the final datasets are in Table I. The source dataset Bili_2M contains 144,146 raw images with an average resolution of 1920x1080. All downstream Bili_* datasets have the same resolution, and the four cross-platform datasets have at least 300x400 resolution, meeting the basic requirements of popular vision encoders. The average word length of all these datasets falls within the range of 16–34.

Other statistics of the NineRec dataset are given in Fig. 2. First, we can see the item distributions of all datasets typically follow the long-tail distribution, which is widely observed in much prior literature [24]. Second, we can see that the number of user interactions are mainly in the range of [5,100], where [5,20] is the majority. We therefore run TransRec experiments by setting the maximum user sequence length to 23 and padding with zeros when user interaction is insufficient. Third, the interactions of Bili_2M occurred mainly between 2017 and 2022.

C. Copyrights and Privacy

In this paper, we strictly adhere to privacy protection measures by collecting only public user behaviors. We have not collected any private user behaviors such as clicks or watching time. Furthermore, the item content we collected, including thumbnails and descriptive texts, is itself freely accessible on the platform's webpages without any limitations. User account IDs and item IDs are also publicly displayed on these platforms. Despite that, we have taken precautions to anonymize them to mitigate potential attacks. These anonymous item IDs can be used to construct item URLs using our mapping algorithm. We

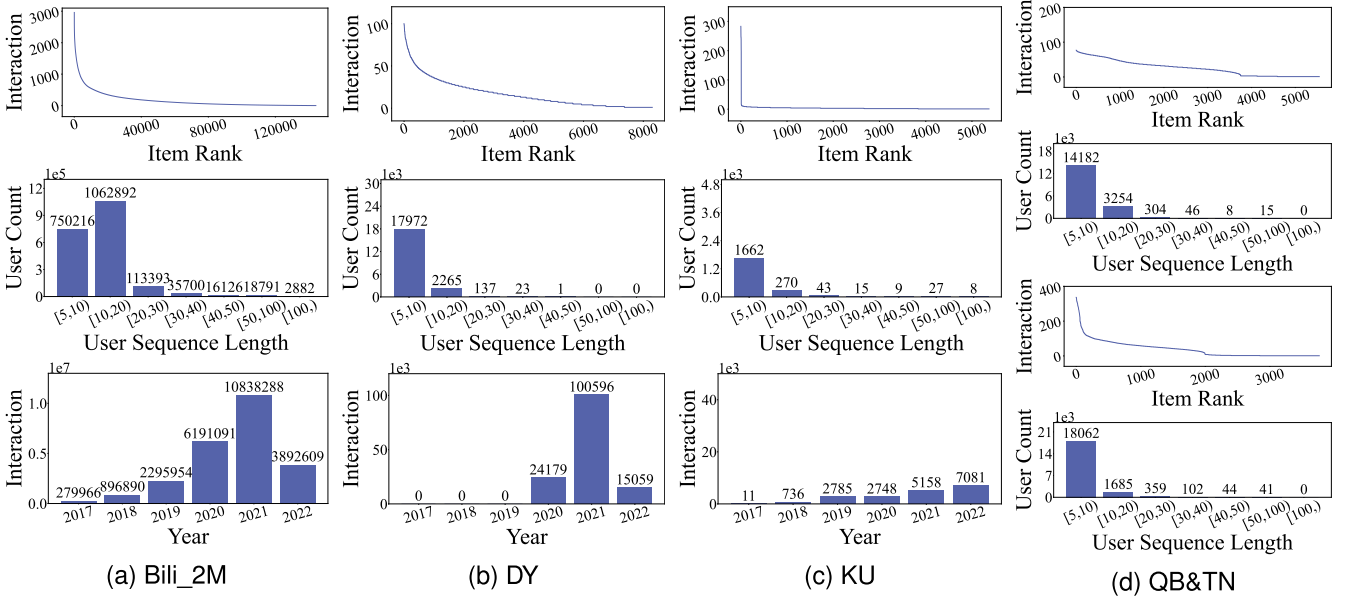


Fig. 2. Dataset details. Top: item popularity distribution; Middle: user interaction length distribution; Bottom: the occurring time of user-item interactions. See more in Appendix Fig. 2, available online.

have implemented the mapping algorithm and URL construction within our download software, ensuring that researchers can only access the data through our downloader.

Regarding copyright concerns, we do not directly provide item cover images. Instead, we offer the downloader tool that allows data users to directly download content from the respective platforms by parsing the provided URLs (the URLs are embedded in the downloader and are not exposed to the public). This approach ensures that copyright issues are not involved and is a widely adopted practice in academic literature [25], [26]. Additionally, for videos that may be expired or unavailable, our downloader automatically locates them in a backup directory to ensure permanent access and downloadability.

D. Comparison to Existing Datasets

The datasets used for the TransRec research can be categorized into three types: datasets with overlapped categorical IDs [7], [24], [27], datasets with pre-extracted features by multimodal encoder [28], [29], [30], [31], [32], and datasets with raw modality features. While there are numerous public datasets available for the former two types, there are very few for the latter type. MIND (for text RS), Amazon (for product RS), Pinterest⁸ [33] (for image RS), Wikimedia [34] (for image RS), GEST & Yelp (for food recommendation), have raw modality features. Among them, MIND, Amazon, Yelp and GEST (a.k.a. Google Restaurants) have a large scale. However, MIND does not have downstream datasets. Though items in Amazon have category information, they are more like cross-category recommendation rather than a strict cross-domain recommendation as there is

not a clear concept of domain [35].⁹ By contrast, NineRec enables both cross-domain and cross-platform recommendation as the target data of NineRec is collected from either different recommendation channels or different systems. Detailed comparison with related datasets is shown in Appendix Table 10, available online.

Another drawback of Amazon is that its images are mainly about single products (e.g. shoes, books, food, electronic products), so models trained on them cannot reflect the real performance in other more complex and practical image scenarios (see Fig. 1). Similarly, Yelp and GEST also suffer from the image diversity issue since most item images are about food and restaurants.

Novelty and Limitations: First, while there are several large-scale public datasets available with raw modality features, their visual or semantic diversity is limited, making them unsuitable as pre-training datasets. For an ideal pre-training model, it is crucial to learn from data with good diversity. In contrast, the source dataset of NineRec contains items from over 20 different video channels, providing a much wider range of visual diversity. In addition, NineRec provides nine target tasks which support both cross-domain and cross-platform recommendation tasks. Second, user behavior observed in existing datasets, such as Amazon and GEST, is not primarily driven by item appearance or modality features. Instead, it is influenced by a myriad of other significant factors, including price, sales, brand, location, and the user's actual purchase needs. That is, user preference cannot be mostly learned from visual features. For instance, when a user purchases baby milk powder on Amazon, it is more

⁸The final released version of Pinterest has only 46,000 users in total and 37,000 items (more than 10 clicks) and no timestamps.

⁹In the Amazon dataset, recommendations for different categories of products are likely to be based on a unified recommendation algorithm. There are some differences compared to true cross-domain or cross-platform recommendation in the strict sense.

likely due to the quality and brand of the product rather than its image characteristics. In contrast, the appearance features in our NineRec dataset which are all collected from content-sharing platforms are *reasonably more important* signals to attract user viewing or clicking actions. Specifically, in the context of short videos and information streams, users tend to passively accept recommendations from the platform, rather than having a specific intent as seen in e-commerce scenarios. Furthermore, a user's decision of whether to click or watch a video is intuitively influenced by the attractiveness of the thumbnail and title. Therefore, from this perspective, NineRec is a more ideal dataset for conducting pure modality feature based recommendation research. Besides, we believe that the recommender systems community requires datasets not only from e-commerce scenarios but also from short videos and information stream contexts. These are highly distinct application domains and it is crucial to take them into consideration when developing recommendation algorithms.

It is worth noting that NineRec also has some limitations: (1) certain user interactions may be influenced by clickbait video thumbnails and titles; (2) the NineRec dataset is sourced from real-world recommendation platforms, resulting in a data distribution that may contain exposure and popularity biases. These factors can potentially impact the fairness of recommender systems. However, we have retained the original data distribution of NineRec to foster research on diversity to the greatest extent possible.

III. RELATED WORK OF TRANSREC

Foundation models [5], trained on broad data at scale and adaptable to a diversity of downstream tasks, have shifted the research paradigm of the AI community from task-specific models to general-purpose models. A broad spectrum of foundation models have been developed in recent years. Amongst them, BERT, RoBERTa, GPT [36], [37], [38], and ChatGPT¹⁰ are renowned for encoding and generating textual data, ResNet, ViT, Swin Transformer [39] and various diffusion models [40] are known for encoding and generating visual data, while CLIP [41] and DALL.E [14] are known for the multimodal research.

Unlike NLP and CV, so far, there are no highly recognized pioneering work on foundation models in the RS community. Recent work such as PeterRec [6], DUPN [42], STAR [43] and Conure [7] have made some meaningful explorations in learning universal (user or item) representation. However, they all belong to the IDRec category, which has limited transfer learning capabilities when the downstream dataset lacks overlapping userIDs or itemIDs [44]. More recently, researchers started to learn the RS models directly from the raw modality features [12], [45]. ZESREC [46] is the first paper that achieved the zero-shot transfer learning ability for text RS without using user or item overlapping information. Similar work includes ShopperBERT [47], PTUM [48], UniSRec [13], IDA-SR [49], VQ-Rec [50], LLM4Rec [16]. All these work focused only on text modality and mainly based on pre-extracted textual features

from a frozen text encoder. Three preprints, i.e. TransRec [15], AdapterRec [10], LLM-REC [51] and concurrent works RecFormer [45] and LMRec [52] performed joint or end-to-end (E2E) training of UE and ME, whereas it remains unknown for other more advanced UE and ME, and training manners. P5 [53], M6-Rec [21] and Conure [7] proposed a unified model to serve multiple tasks, such as review summary, rating prediction, user profile prediction, and item recommendation.

In this paper, we present benchmark results on E2E-learned TransRec, which is computationally expensive but performs much better than pre-extracted features.

IV. BASELINES OVERVIEW

Modality-Based Recommendation (MoRec): Let $\mathcal{U}, \mathcal{I}$ be the set of users and items respectively. The goal of RS is to predict the potential interaction of user $u \in \mathcal{U}$ by exploiting her past behaviors $\mathcal{I}_u = \{i_1, \dots, i_n\}$. In a classical IDRec setup, users and items are usually represented by their unique IDs. Accordingly, the userIDs and itemIDs can be embedded into a series of dense vectors, denoted as $\beta_u \in R^d$ and $\beta_i \in R^d$, where d is the embedding size, and each of them is the representation of a user or item. MoRec instead applies an modality encoder (ME), denoted as $f(x_i)$, to encode xmodality features x_i of an item i . MoRec can basically inherit other modules in IDRec, such as the user encoder or recommendation backbone. In theory, various MoRec models can be constructed by simply replacing the β_i of IDRec with $f(x_i)$. In this paper, we limit the scope of MoRec to learn recommendation models only from pure modality features instead of treating them as auxiliary features of ID features. This distinguishes from a majority of previous works [28], [54] using ID as the main feature and modality as the side feature. However, such paradigms are not well-suited to achieve the goal of transferable recommendation due to the practical challenges associated with sharing or transferring ID features [12], [13], [15].

TransRec: A RS model is usually composed of a user encoder (UE) $g(x_u)$, item encoder $f(x_i)$ and their dot product $\hat{y} = g(x_u) \otimes f(x_i)$.¹¹ To realize a foundation TransRec model, both UE and ME should be transferable. That is, the widely used userID should not exist in TransRec either. The common approach is to replace userID by a sequence of her interacted items $\mathcal{I}_u$, which are again encoded by ME [46], that is, $g(x_u) = G(f(x_{i_1}), \dots, f(x_{i_n}))$ where $G(\cdot)$ can usually be a sequential encoder. In view of this, existing TransRec models are mainly sequential recommendation models or sequential MoRec, e.g. PTUM [48], CLUE [55], TransRec [15], UniSRec [13], VQ-Rec [50] and AdapterRec [10]. In this paper, we benchmark TransRec using the most well-known $G(\cdot)$, including *RNN*-based GRU4Rec [56], *CNN*-based NextItNet, multi-head self-attention (*MHSA*) based SASRec, BERT4Rec, and a standard *DNN*-based encoder. While there are some new SOTA sequential models in literature, we find that most of them

¹⁰[Online]. Available: <https://openai.com/blog/chatgpt/>

¹¹On top of $f(x_i)$ there are usually one or more DNN layers for dimensionality transformation. For simplicity, we omit related formulas.

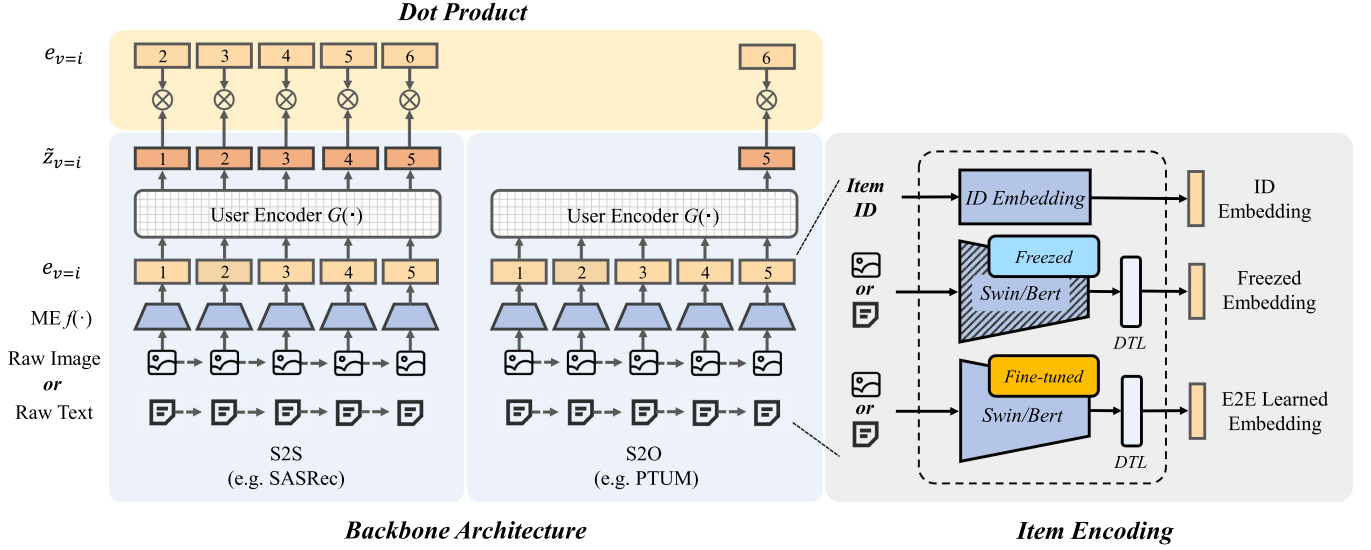


Fig. 3. TransRec architectures (S2S and S2O). BERT and Swin-B (Swin Transformer) are used as ME. DTL is the DNN layers for dimension transformation. UE can be a stack of DNN, RNN, CNN, or MHSA layers. $\tilde{z}_{v=1}, \dots, \tilde{z}_{v=n}$ are vector generated by UE, $e_{v=1}, \dots, e_{v=n+1}$ are vectors generated by ME.

can be seen as variants of the above classic models (especially a variant of Transformer [57]).

Training Details: The TransRec model will first be pre-trained with sufficient data in the source domain, and then fine-tuned to serve various target domains with relatively less data. The training process of TransRec has no big difference from IDRec models. It involves computing the dot product(s) of user embeddings and item embeddings for both a positive user-item pair and a randomly selected negative pair. Subsequently, the typical binary cross-entropy loss is calculated based on these dot products. Recent literature [12], [45], [58] clearly demonstrates that end-to-end (E2E) learning is significantly more effective compared to using pre-extracted modality features from a frozen multimodal encoder. Therefore, in our baselines, we adopt E2E learning to report baseline results.

Second, we evaluate two popular training modes: sequence-to-sequence (S2S) and sequence-to-one (S2O), see Fig. 3. They both encode a sequence of items as input, S2O predicts only the last item while S2S predicts a sequence of items. That is, the (input $\rightarrow$ output) format of S2S is $i_1, i_2, \dots, i_{n-1} \rightarrow i_2, i_3, \dots, i_n$, and format of S2O is $i_1, i_2, \dots, i_{n-1} \rightarrow i_n$. Clearly, the S2O training architecture is essentially a variant of two-tower DSSM [59] model, where one tower represents user sequence and the other represents target item. In this paper, we optimize all parameters on both the source and target datasets. In practice, comparable results may be obtained by tuning a few top layers for some datasets.

V. TRANSREC BENCHMARK

A. Evaluation

We adopt the leave-one-out strategy to split each dataset, namely, the last interaction per user is used for testing, the second to last is used for validation, and the rest are used for training.

The popular H@10 (Hit Ratio @10) and N@10 (Normalized Discounted Cumulative Gain @10) are used as the evaluation metrics [24]. To save space, we report results of N@10 in Appendix, available online. We rank the predicted item among all items in the pool instead of drawing 100 random items [60].

B. Experimental Setting

Considering the early stage of TransRec¹², it is crucial to conduct a fair comparison between TransRec and the well-established and dominant IDRec models. To ensure fairness, we make sure that both IDRec and TransRec employ the same network backbone and training approach. This includes using identical loss functions and samplers, with the only difference being the item encoder, which is replaced with a state-of-the-art modality encoder in TransRec. This setup enables a fair and direct comparison between the two models. Some literature utilized relatively smaller ID embedding sizes for IDRec, making their MoRec or TransRec easier to achieve improvements in performance. Additionally, there are also studies that compare TransRec and IDRec using different network backbones and samplers. However, we believe that conducting a fair comparison between the two models becomes challenging when multiple factors differ between them.¹³

Regarding the hyper-parameter setting, our first principle is to ensure that IDRec on both upstream and downstream datasets are *extremely* tuned, including learning rate γ , embedding/hidden size d , layer number l , dropout ρ , batch size b , etc. For example,

¹²In fact, the community is still unable to provide a definitive answer regarding the possibility of developing a one-for-all foundation model for recommender systems.

¹³Repeatability is a growing concern in the recommender system community. The absence of a public and community-assessed benchmark and leaderboard creates difficulties in accurately assessing the real progress made in the field, see [61], [62], [63], [64].

TABLE II
TRANSREC RESULTS (%) ON DOWNSTREAM DATASETS

Dataset	Metric	SASRec			BERT4Rec			NextItNet			GRU4Rec		
		IDRec	NoPT	HasPT	IDRec	NoPT	HasPT	IDRec	NoPT	HasPT	IDRec	NoPT	HasPT
BERT (base version) for text recommendation													
Bili_Food	H@10	18.09	18.03	<u>19.59</u>	17.07	16.70	<u>20.00</u>	11.75	15.80	<u>17.26</u>	11.49	16.12	<u>17.21</u>
Bili_Dance	H@10	23.46	23.49	<u>25.41</u>	21.92	22.14	<u>26.84</u>	17.52	21.11	<u>23.33</u>	17.38	21.25	<u>22.47</u>
Bili_Movie	H@10	11.40	11.64	<u>12.63</u>	10.75	11.12	<u>13.47</u>	7.95	10.34	<u>11.57</u>	6.85	9.67	<u>11.70</u>
Bili_Cartoon	H@10	11.63	11.94	<u>13.75</u>	11.54	12.14	<u>14.01</u>	8.17	11.07	<u>11.85</u>	8.73	11.04	<u>12.69</u>
Bili_Music	H@10	19.16	19.42	<u>20.65</u>	17.73	17.44	<u>20.95</u>	14.69	17.84	<u>19.29</u>	15.52	16.19	<u>17.79</u>
KU	H@10	28.36	30.77	<u>31.36</u>	24.18	24.13	<u>29.30</u>	22.22	27.38	<u>27.92</u>	20.40	28.61	<u>29.60</u>
QB	H@10	33.92	34.27	<u>34.60</u>	32.80	32.28	<u>33.40</u>	<u>31.77</u>	30.28	29.98	30.75	32.82	<u>33.24</u>
TN	H@10	15.74	15.11	<u>16.85</u>	15.77	15.14	<u>16.86</u>	<u>12.86</u>	12.79	11.96	14.16	<u>15.73</u>	14.01
DY	H@10	<u>15.92</u>	14.35	<u>14.49</u>	13.43	9.72	<u>13.60</u>	<u>12.06</u>	10.78	9.51	10.45	16.24	<u>16.34</u>
Swin Transformer (base version) for image recommendation													
Bili_Food	H@10	18.09	17.20	<u>18.72</u>	17.07	2.15	<u>19.01</u>	11.75	14.52	<u>17.89</u>	11.49	16.33	<u>17.74</u>
Bili_Dance	H@10	<u>23.46</u>	21.63	<u>22.16</u>	<u>21.92</u>	16.65	<u>20.90</u>	17.52	17.12	<u>21.92</u>	17.38	19.56	<u>21.84</u>
Bili_Movie	H@10	11.40	10.30	<u>11.50</u>	<u>10.75</u>	1.64	<u>10.11</u>	7.95	8.21	<u>10.73</u>	6.85	8.99	<u>10.44</u>
Bili_Cartoon	H@10	11.63	11.09	<u>11.78</u>	11.54	10.54	<u>11.82</u>	8.17	9.00	<u>10.92</u>	8.73	8.95	<u>10.48</u>
Bili_Music	H@10	<u>19.16</u>	17.17	<u>17.56</u>	<u>17.73</u>	11.92	<u>15.42</u>	14.69	15.58	<u>16.76</u>	15.52	15.20	<u>16.04</u>
KU	H@10	28.36	<u>33.08</u>	33.03	24.18	23.75	<u>25.42</u>	22.22	27.23	<u>32.64</u>	20.40	<u>31.36</u>	30.18
QB	H@10	<u>33.92</u>	32.39	<u>33.57</u>	<u>32.80</u>	25.96	22.33	<u>31.77</u>	29.91	31.75	30.75	32.53	<u>33.04</u>
TN	H@10	15.74	14.12	<u>14.44</u>	<u>15.77</u>	13.59	12.98	<u>12.86</u>	11.40	<u>12.15</u>	14.16	<u>14.27</u>	13.81
DY	H@10	<u>15.92</u>	14.08	<u>14.68</u>	<u>13.43</u>	10.83	9.40	12.06	11.60	<u>12.49</u>	10.45	13.26	<u>13.93</u>

NoPT means that it is directly trained on the dataset without pre-training (PT) on the source or other RS data (Note ME in NoPT was pre-trained on NLP or CV data.). HasPT means it has been pre-trained on the source dataset and then fine-tuned on the target dataset. The NDCG@10 results are given in Appendix Table 1. Results in italics denote model collapse. Results in bold indicate the maximum between NoPT & HasPT. Underlined results are the maximum value among all.

we tune γ by searching from [5e-6, 1e-5, 5e-5, 1e-4, 5e-4, 1e-3], d from [64, 128, 256, 512, 1024, 2048]. Similarly, we find optimal values for other hyperparameters. While for TransRec, we first use the same set of hyper-parameters as IDRec and then perform the search around the best choices (the search range and step size are kept exactly the same as IDRec). This is a faster and fair way to find better hyper-parameters for TransRec on both source and target datasets. It is worth noting that iterating over all hyper-parameter combinations for TransRec is infeasible since training it usually takes 100x larger compute and time than IDRec by the E2E manner (see details in Appendix Table 5, available online).

All images in NineRec are resized to the shape of 224×224 pixels. The text descriptions are limited to a maximum of 30 Chinese/English words.

C. Benchmarking User Encoders

In Table II, our benchmark covers several most classical recommendation backbones (RNN-based GRU4Rec, CNN-based NextItNet, MHSA-based SASRec and BERT4Rec, and two DNN-based models in Appendix Table 6, available online), trained end-to-end on *two* single modalities in *nine* target tasks, by replacing their original itemID with item ME. We also report two additional peer-reviewed baseline UniSRec [13] and VQ-Rec [50] in Appendix Table 11. Our results indicate that these models do not outperform the classical methods under the fair comparison setting.

Regarding the modality encoder, we use BERT¹⁴ for text recommendation, and use Swin Transformer for image recommendation. Without special mention, all models here are trained using the S2S mode (see Fig. 3). In addition, (1) we report the results of multimodal recommendation with the SASRec backbone in Table VI and Appendix Table 9, available online; (2) we report two DNN backbone baselines in Appendix Table 6, available online; (3) we report the S2O training baseline in Table VII; (4) we report the results of the source Bili_500K dataset in Appendix Table 7, available online; (5) We report results on the source and target datasets using the larger Bili_2M dataset in Appendix Tables 2 and 8, available online.

Beyond the benchmark results, we show some insightful findings as below. Note in this paper, we mainly use Bili_500K as the source dataset unless otherwise stated, and report some key results using Bili_2M given the ultra-high training cost.

- Tables II, VI, VII, and Appendix Table 6, available online show that TransRec pre-trained on the source dataset (i.e. HasPT) mostly performs better than its NoPT version. These results highlight the effectiveness of pre-training and indicate that the NineRec dataset is well-suited for research on transfer learning.

¹⁴In this paper, we use the Chinese BERT as ME, while due to lack of other Chinese text ME, we use English ME by translating the text into English for RoBERTa, OPT, CLIP and ViLT. Details are provided in Appendix Table 4, available online.

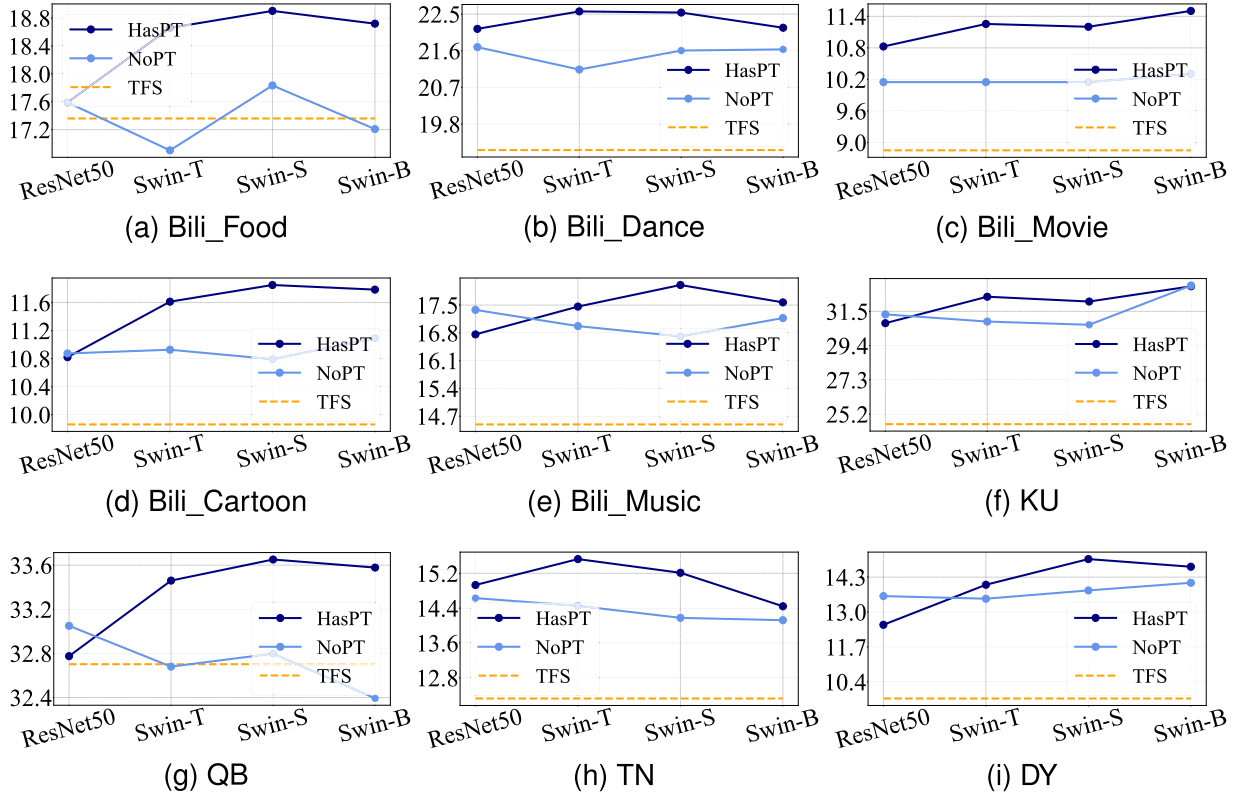


Fig. 4. Benchmark results (y -axis: %) of item ME (with SASRec as UE). The details of ResNet50, Swin-T, Swin-S and Swin-B are provided in Appendix Table 4, available online. All hyper-parameters are kept the same for NoPT, HasPT, and TFS. TFS means TransRec is not pre-trained on the source dataset, and its ME is not pre-trained on ImageNet. The dashed yellow line only shows ResNet50.

- Tables II and VII show TransRec pre-trained on text modality in general obviously outperforms its IDRec counterpart on these downstream datasets, meanwhile, it sometimes performs worse than IDRec if pre-trained on image modality. Similar results can be even observed on the source datasets and two very warm datasets in Appendix Tables 7 and 8, available online. Previous works have primarily focused on TransRec beating IDRec in cold-start scenarios. However, defeating IDRec in non-cold-start scenarios signifies a significant advancement and potentially heralds a paradigm shift in the future of recommender systems. This is particularly noteworthy considering that IDRec has remained the state-of-the-art approach for over 10 years.
- Table VI shows that TransRec models trained on multi-modal (text and image) features do not consistently outperform those trained on a single modality (i.e. Table II). This is a reasonable observation (also observed in literature [65]), as effectively fusing text and image modalities in recommendation models poses a non-trivial challenge that remains largely unexplored within the E2E learning paradigm.
- Table II and Appendix Table 7, available online indicate that a recommendation network with higher accuracy on IDRec, such as SASRec compared to BERT4Rec, may not necessarily result in higher accuracy on TransRec or MoRec, even when utilizing the same ME.

A surprising result is that model collapse may happen during learning MoRec/TransRec, as shown in Table II marked with italics. We find that it is quite difficult to jointly learn BERT4Rec with Swin Transformer sometimes even many hyper-parameter searches are performed. This is unknown to the community. More interesting findings can be made according to such extensive results (see Appendix, available online).

D. Benchmarking Item Encoders

Fig. 4 and Table V present the evaluation of several well-known item ME models, such as ResNet and Swin Transformer with different model sizes, for image recommendation, and RoBERTa and OPT [66] for text recommendation. Unless otherwise specified, the SASRec backbone and S2S training mode are utilized by TransRec in the subsequent sections. The majority of observations align with the above findings. Interestingly, we discover that pre-training TransRec on the source Bili_500K dataset (HasPT), using ResNet50 as the ME, does not consistently yield superior outcomes compared to its NoPT version. This outcome is somewhat unexpected, as it suggests that the parameters of ResNet50 become degraded after pre-training on Bili_500K. This is unexpected but is not impossible. In fact, Table II also showed several similar results. To see why, we show the results of ResNet50 without pre-training on ImageNet (i.e. TFS). It can be clearly seen that TFS largely underperforms both NoPT and HasPT. This indicates that ResNet50 parameters

TABLE III
E2E vs TS ON HR@10 (%)

Dataset	BERT			Swin-B		
	TS	E2E	Improv.	TS	E2E	Improv.
Bili_500K	0.57	4.69	8.22x	0.41	3.57	8.71x
Bili_Food	11.52	19.59	1.7x	14.20	18.72	1.32x
Bili_Dance	14.32	25.41	1.77x	17.57	22.16	1.26x
Bili_Movie	5.73	12.63	2.2x	7.01	11.50	1.64x
Bili_Cartoon	6.28	13.75	2.19x	6.70	11.78	1.76x
Bili_Music	11.01	20.65	1.88x	12.83	17.56	1.37x
KU	27.00	31.36	1.16x	30.33	33.03	1.09x
QB	28.14	34.60	1.23x	29.75	33.57	1.29x
TN	8.28	16.85	2.04x	10.37	14.44	1.39x
DY	6.90	14.49	2.1x	10.27	14.68	1.43x

TS means the parameters of ME (pre-trained on NLP or CV data) are not allowed to be optimized during training on both the source and target datasets.

pre-trained on ImageNet are highly beneficial as an initialization step, and as a result, additional training on Bili_500K may not always bring significant benefit for other downstream tasks.

E. End-to-End (E2E) Versus Two-Stage (TS) Benchmark

By surveying the literature, we found that most previous MoRec/TransRec studies adopt a two-stage (TS) training approach [67], [68], [69]: first pre-extract offline modality features via ME, and then incorporate them into the recommendation model as regular features. In recent two years, the E2E training approach has attracted attention, but mainly for text recommendation [45], [58]. We report the results of E2E versus TS in Table III. Clearly, TransRec by E2E training of ME outperforms the TS method substantially on both text and image modalities. For some text recommendation tasks (e.g. Bili_Movie, Bili_Cartoon, TN, and DY), E2E could achieve about more than 200% higher accuracy. The results indicate that the off-the-shelf representation features extracted directly from the pre-trained modality encoders have a considerable gap between the NLP, CV and recommendation tasks, i.e. these features are not universal or at least not specific enough to the recommendation task. Parameter retraining on the target datasets is a key way to obtain desired results. Thereby, NineRec with raw text and image features will serve as an important dataset for studying E2E-learning based MoRec and TransRec, although the computational cost is high (see Appendix Table 5, available online).

VI. ZERO-SHOT RECOMMENDATION

Zero-shot learning is a very challenging task in NLP and CV. Although pre-trained TransRec achieves competitive results via fine-tuning, we ideally want it to achieve satisfactory results without parameter fine-tuning on the downstream dataset. This is also an important goal of foundation models. We refer to such recommendation setting as zero-shot recommendation in agreement with [14].

We report results in Table IV. First, we can see that the TransRec model after pre-training (but without fine-tuning) can achieve 7x-70x (e.g. 0.16vs. 11.32) better results than the

TABLE IV
RESULTS ON HR@10 (%) FOR ZERO-SHOT TEXT RECOMMENDATION
(APPENDIX TABLE 3, AVAILABLE ONLINE FOR IMAGE RECOMMENDATION)

Type	Bili_Food	Bili_Music	KU	QB	DY
Random	0.63	0.16	0.49	0.18	0.12
ZeroRec	4.76	11.30	4.96	7.58	0.88
	Bili_Dance	Bili_Movie	Bili_Cartoon	TN	
Random	0.43	0.28	0.21	0.26	
ZeroRec	8.30	5.16	5.95	1.21	

ZeroRec is TransRec that is pre-trained on the source dataset, and then directly predicts on the target datasets. Random baseline is equivalent to the accuracy of IDRec in the new item setting.

TABLE V
RESULTS OF MORE TEXT ME (WITH SASREC AS UE)

Dataset	RoBERTa _{base}			OPT _{125M}		
	NoPT	HasPT	Improv.	NoPT	HasPT	Improv.
Bili_500K	3.82 2.00	- -	- -	3.48 1.82	- -	- -
Bili_Food	17.94 9.52	18.08 10.18	+0.77% +6.48%	17.29 9.76	18.67 10.69	+7.39% +8.70%
Bili_Dance	22.97 13.00	23.76 13.56	+3.32% +4.13%	22.56 12.99	23.80 13.38	+5.21% +2.91%
Bili_Movie	11.18 5.95	12.43 6.58	+10.06% +9.57%	11.24 6.12	12.25 6.53	+8.24% +6.28%
Bili_Cartoon	11.77 6.48	12.44 6.74	+5.39% +3.86%	11.75 6.53	12.51 6.92	+6.08% +5.64%
Bili_Music	18.63 10.72	19.47 11.07	+4.31% +3.16%	18.59 10.73	19.12 10.85	+2.77% +1.11%
KU	29.11 24.34	30.92 25.88	+5.85% +5.95%	28.71 24.81	30.73 25.25	+6.57% +1.74%
QB	34.11 26.26	33.62 26.38	-1.46% +0.45%	33.02 25.59	33.28 26.49	+0.78% +3.40%
TN	15.68 8.76	14.89 8.64	-5.31% -1.39%	14.40 8.00	14.40 8.19	+0.00% +2.32%
DY	13.97 7.50	14.29 8.14	+2.24% +7.86%	13.28 7.39	13.65 7.82	+2.71% +5.50%

The upper results denote HR@10, the lower denotes NDCG@10. Note that since RoBERTa and OPT have no Chinese version, we translated the Chinese text into English (by DeepL: <https://www.deepl.com/translatorand>) then performed the evaluation. English translation will be provided with the datasets. Again, NoPT means that it is directly trained on the dataset without pre-training (PT) on the source or other RS dataset (Note ME in NoPT was pre-trained on NLP or CV data.). HasPT means it has been pre-trained on the source dataset and then fine-tuned on the target.

random baseline. This clearly shows that the pre-trained representations in the source domain have some generality. Second, we also find that TransRec's zero-shot prediction performance is far behind its fine-tuning method (see Table II). This suggests that the pre-trained representations in the source domain are far from perfect. We speculate it might have improved performance by pre-training on a significantly larger source dataset (e.g. 100x-1000x larger) or multiple diverse source datasets with a larger model size. This phenomenon is known as the emergent abilities of foundation models [70]. In other words, like NLP and CV, recommendation models also face great challenges on zero-shot tasks. We are unsure whether NineRec could be used to address this issue, but we believe NineRec will inspire new work and new datasets.

TABLE VI
RESULTS OF MULTIMODAL RECOMMENDER SYSTEM ON THE TARGET DATASETS

Dataset	Metric	BERT+Swin-T		
		NoPT	HasPT	Improv.
Bili_Food	H@10	16.68	17.36	+3.92%
	N@10	9.05	9.51	+4.84%
Bili_Dance	H@10	20.74	24.03	+13.69%
	N@10	11.60	13.62	+14.83%
Bili_Movie	H@10	10.27	12.54	+18.10%
	N@10	5.46	6.47	+15.61%
Bili_Cartoon	H@10	10.32	12.35	+16.44%
	N@10	5.19	6.72	+22.77%
Bili_Music	H@10	16.93	18.84	+10.14%
	N@10	9.52	10.58	+10.02%
KU	H@10	28.96	29.60	+2.16%
	N@10	23.80	24.67	+3.53%
QB	H@10	31.52	33.15	+4.91%
	N@10	22.53	24.15	+6.71%
TN	H@10	14.88	15.53	+4.19%
	N@10	8.14	9.00	+9.56%
DY	H@10	11.29	12.90	+12.48%
	N@10	5.87	6.92	+22.77%

Given much worse results of CLIP and ViLT on the source dataset (see Appendix Table 9), we only evaluate BERT+Swin-T as ME here. SASRec is used as UE.

VII. CONCLUSION

Developing a research direction in the field of recommender systems (RS) is challenging without access to large-scale and real-world datasets; similarly, measuring genuine progress without a public benchmark is equally difficult. In this paper, we present the NineRec dataset suite and benchmarks, which are designed to advance transfer learning and foundation models in the RS field by leveraging raw and pure modality features. Through empirical studies, we also report several noteworthy findings. Given the rapid advancements in the field and the high computational demands, it is not feasible to evaluate all existing RS architectures, variants, and settings (such as various samplers and loss functions). However, we can establish public leaderboards to facilitate tracking of the latest state-of-the-art models by the community.

There are many limitations and challenges not addressed in this paper. First, while we have developed TransRec by assembling popular user encoder (UE) from IDRec and popular item modality encoder (ME) from NLP and CV, we acknowledge that this may not be the optimal approach. It is possible that only specifically designed UE and ME can fully realize the transfer learning potential of TransRec. Second, we need to consider whether the optimization and hyper-parameter search techniques developed for IDRec over the past decade are also applicable to MoRec and TransRec. Third, we need to investigate the proper alignment and fusion of multimodal features within the end-to-end learning paradigm. In addition, we need to address the significant computational costs associated with end-to-end training TransRec in practical systems. This is particularly crucial when dealing with datasets that are 100x or 1000x larger than the ones used in this study. In reality, TransRec or *foundation* models are still in the early stages of development for recommendation problems. To date, there is no widely recognized TransRec paradigm. However, we believe that NineRec can help

TABLE VII
RESULTS OF S2O FRAMEWORK (SEE FIG. 3) WITH THE MHSA OR SASREC BACKBONE

Dataset	IDRec	BERT			Swin-B		
		NoPT	HasPT	Improv.	NoPT	HasPT	Improv.
Bili_500K	0.87	2.48	-	-	1.75	-	-
	0.45	1.31	-	-	0.87	-	-
Bili_Food	11.91	15.36	16.58	+7.37%	14.44	14.97	+3.57%
	7.14	8.18	9.10	+10.06%	7.95	8.15	+2.47%
Bili_Dance	18.15	20.02	22.00	+8.99%	17.49	19.59	+10.71%
	11.09	11.56	12.65	+8.65%	9.49	10.88	+12.83%
Bili_Movie	7.68	9.36	10.05	+6.86%	8.15	8.65	+5.80%
	4.51	5.21	5.32	+2.06%	4.41	4.43	+0.41%
Bili_Cartoon	8.51	9.89	11.18	+11.57%	6.11	9.31	+1.91%
	4.95	5.32	6.11	+12.97%	5.13	4.98	-2.97%
Bili_Music	16.68	17.12	17.62	+2.83%	15.26	15.27	+0.12%
	10.16	9.51	10.06	+5.56%	8.44	8.61	+1.92%
KU	20.74	26.59	26.84	+0.92%	28.26	26.74	-5.39%
	19.40	23.38	23.38	+0.00%	23.77	23.31	-1.93%
QB	27.03	28.78	29.13	+1.21%	28.09	28.51	+1.46%
	23.94	22.01	22.47	+2.03%	21.28	23.16	+8.09%
TN	11.62	12.74	12.73	-0.08%	11.79	12.38	+4.74%
	7.16	7.41	6.92	-6.57%	6.69	7.11	+5.89%
DY	13.84	12.11	10.95	-9.59%	11.97	12.19	+1.81%
	8.99	7.25	6.09	-15.95%	6.97	7.13	+2.25%

The upper results denote HR@10, while the lower denotes NDCG@10. The only difference of S2O training and DSSMvariant (see Appendix Figure 3) is that S2O uses the MHSA layers as the backbone but DSSM-variant uses the DNN layers as the backbone. We can see that the S2O training mode is very worse than the S2S model (Table 2) even they both use MHSA layers as the backbone. ‘-’ means that there is no pre-training stage on the source dataset.

advance the field by inspiring new questions, new ideas, and new research.

In this paper, we primarily study NineRec for transferable recommendation research. However, there are several other potential applications of NineRec in the RS field. For instance, many widely-used RS datasets only provide itemID information, which limits researchers’ ability to fully understand what their recommender systems are recommending, beyond an accuracy score. By utilizing NineRec, researchers can gain a better understanding of their RS models, particularly for interpretable RS [71] and visually-aware RS evaluation [72] problems. This can ultimately lead to more effective and explainable RS models. Furthermore, many researchers in the NLP and CV fields are currently working on developing modality encoders with universal representations [73]. However, these models are often only evaluated on standard NLP and CV tasks, such as image classification. We contend that the recommendation task, which involves predicting user preferences, is more challenging than these basic downstream tasks. Thus, NineRec could be crucial for NLP and CV researchers and may even facilitate the integration of RS with the NLP and CV fields.

VIII. AUTHOR CONTRIBUTIONS

Fajie is the corresponding author who designed, supervised and funded this research; Jiaqi performed this research, in charge of technical parts; Fajie, Jiaqi wrote the paper and answered the reviewers’ questions; Chengyu, Zheng, Youhua assisted some key experiments; Yunzhu, Yongxin, Jie together collected the 10 datasets.

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